AI Model Card

AccuContour

Company: Manteia | Generated: 10/23/2026

General Information

Product Name:	AccuContour
Company:	Manteia
Version:	4.0
Category:	Auto-Contouring
Release Date:	1/23/2026
Last Updated:	9/29/2026
Description:	AccuContour 4.0 is used by the radiation oncology department to segment CT images and generate information needed for treatment planning, treatment evaluation, and treatment adaption. (Source: FDA 510(k) K251351 Summary, accessed 2026-06-11.)

Evidence Classification

Rigor E1 — Exploratory or single-center technical validation

Efficacy L3 — Decision/workflow efficacy

Readiness R3 — Enhanced implementation readiness

+ Vendor-independent

- Multi-centre

- Multi-national

- Prospective

+ External validation

9 scored publication(s) · 9 listed · score origin: papers
Rigor: Wang 2022 (Front Oncol) confirmed; additional independent studies support E2 (conditional on product-name verification in full text). 2026-09-23 manufacturer material review: Li et al., Radiother Oncol 2022 (doi 10.1016/j.radonc.2022.11.004, DOI resolved via Crossref) added as a scored paper — six abdominal/pelvic patients on an MR-linac, Manteia co-authored, so rigor stays E1.
Efficacy: Evidence supports contouring efficiency and geometric agreement for selected OARs, with clinically necessary review/editing; impact is workflow/contouring support rather than demonstrated patient outcome improvement. Raised to I2 on 2026-09-23: Li et al. 2022 reports measured workflow endpoints (contouring 73.4 ± 6.5 s vs 12–22 min manual; adaptive session 1650 ± 274 s vs 3251.8 ± 447 s) in a six-patient MRgART series. No patient-outcome endpoint is demonstrated.
Readiness: Peer-reviewed retrospective evidence, FDA 510(k) clearance for v4.0 and a manufacturer-supplied MDR certificate listing AccuContour v3.1 are present; local commissioning, model acceptance testing, governance, and user training remain required. The certificate does not establish cleared MRI segmentation or cover the FDA v4.0 version.

Key Features

- AI-powered segmentation
- Fast processing
- Clinical workflow integration
- Guideline-based contouring support

For information only. Not a clinical validation or endorsement; verify with the manufacturer and local commissioning.

- Target and OAR contouring

Technical Specifications

Input Formats:	DICOM
Output Formats:	DICOM-RTSTRUCT
Processing Time:	~1 minute per case (site- and ROI-dependent)
Integration Options:	TPS integration; PACS integration
Deployment Options:	Cloud-based; On-premises
Target Population:	Patients selected for radiation therapy; pediatric performance requires local validation
Modality Support:	CT; MRI
Target Anatomy:	Brain; Head & Neck; Thorax; Abdomen; Pelvis
Disease Targeted:	Multiple Cancer Types

Supported Structures (483)

Head & Neck — 156

A_CommonCarotid_L	A_CommonCarotid_R	A_InternalCarotid_L
A_InternalCarotid_R	Bone_EusTube_L	Bone_EusTube_R
Brain	Brainstem	BuccalMucosa_L
BuccalMucosa_R	Canal_Auditory_L	Canal_Auditory_R
Canal_IA_L	Canal_IA_R	Cavity_Oral
Cavity_Oral_wGum	Cavity_Tymp_L	Cavity_Tymp_R
Cerebellum	Cochlea_L	Cochlea_R
CricoidCartilage	Ear_Internal_L	Ear_Internal_R
Ear_Middle_L	Ear_Middle_R	EyeBall_L
EyeBall_R	Eye_L	Eye_R
FourthVentricle	FrontalSuture	Frontomaxillary
Frontozygomatic	Glnd_Submand_L	Glnd_Submand_R
Glnd_Thyroid	HardPalate	Hippocampus_L
Hippocampus_R	Hyoid	Hypopharynx
Joint_TM_L	Joint_TM_R	Lacrimal_L
Lacrimal_R	Larynx	LarynxAir
LarynxExtend	LarynxGlottic	LarynxSubglottis
LarynxSupraglottic	Lens_L	Lens_R
Lips	Lobe_Temporal_L	Lobe_Temporal_R
Lobe_Temporal_wHip_L	Lobe_Temporal_wHip_R	Mandible
Mandible_L	Mandible_R	Masseter_L
Masseter_R	Mastoid_L	Mastoid_R
Musc_Autochthon_L	Musc_Autochthon_R	Musc_Digastric_L
Musc_Digastric_R	Musc_InferiorOblique_L	Musc_InferiorOblique_R
Musc_InferiorRectus_L	Musc_InferiorRectus_R	Musc_LateralPterygoid_L
Musc_LateralPterygoid_R	Musc_LateralRectus_L	Musc_LateralRectus_R
Musc_LevatorPalpebraeSuperioris_L	Musc_LevatorPalpebraeSuperioris_R	Musc_LevatorScapulae_L
Musc_LevatorScapulae_R	Musc_MedialPterygoid_L	Musc_MedialPterygoid_R
Musc_MedialRectus_L	Musc_MedialRectus_R	Musc_MiddleScalene_L

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Head & Neck — 156 (cont.)

Musc_MiddleScalene_R	Musc_Platysma_L	Musc_Platysma_R
Musc_PosteriorScalene_L	Musc_PosteriorScalene_R	Musc_Prevertebral_L
Musc_Prevertebral_R	Musc_SternoThyroid_L	Musc_SternoThyroid_R
Musc_Sternocleidomastoid_L	Musc_Sternocleidomastoid_R	Musc_SuperiorOblique_L
Musc_SuperiorOblique_R	Musc_SuperiorRectus_L	Musc_SuperiorRectus_R
Musc_Temporalis_L	Musc_Temporalis_R	Musc_Thyrohyoid_L
Musc_Thyrohyoid_R	Musc_Trapezius_L	Musc_Trapezius_R
NasalCavity_L	NasalCavity_R	NasalSeptum
Nasopharynx	OpticChiasm	OpticNrv_L
OpticNrv_R	Oropharynx	Parotid_L
Parotid_R	PharyngealConstrictors_L	PharyngealConstrictors_M
PharyngealConstrictors_U	Pituitary	PosteriorFossa
Process_Condyloid	Process_Coronoid	Skull
SoftPalate	StyloidProcess_L	StyloidProcess_R
ThirdVentricle	ThyroidCartilage	Tongue
V_InternalJugular_L	V_InternalJugular_R	VentricleBody_L
VentricleBody_R	VentricleFrontalHorn_L	VentricleFrontalHorn_R
VentricleOccipitalHorn_L	VentricleOccipitalHorn_R	VentricleTemporalHorn_L
VentricleTemporalHorn_R	VentricleTrigone_L	VentricleTrigone_R
Vertebrae_C1	Vertebrae_C2	Vertebrae_C3
Vertebrae_C4	Vertebrae_C5	Vertebrae_C6
Vertebrae_C7	Vestibule_L	Vestibule_R
ZygomaticArch_L	ZygomaticArch_R	Zygomaticomaxill

Head & Neck LN — 12

III_MidJugular	IIa_UpperJugular	IIb_UpperJugular
IVa_LowerJugular	IVb_MedialSupraclavicular	Ia_Submental
Ib_Submandibular	VIa_AnteriorJugular	VIb_Paratracheal
Va_UpperPosteriorTriangle	Vb_LowerPosteriorTriangle	Vc_LateralSupraclavicular

Thorax — 97

A_Aorta	A_Pulmonary	A_Subclavicle_L
A_Subclavicle_R	Aorta_Ascending	Aorta_Descending
AtrialAppendage_L	BrachialPlex_L	BrachialPlex_R
BrachiocephalicTrunk	Breast_L	Breast_R
BronchialTree	ChestWall_L	ChestWall_R
Clavicle	Clavicle_L	Clavicle_R
CostalCartilages	Esophagus	Fat_Thoracic
GreaterTubercle	Heart	HumeralHead_L_G
Humerus_L	Humerus_R	LA
LAD	LCX	LM
LV	LesserTubercle	LungLowerLobe_L
LungLowerLobe_R	LungMiddleLobe_R	LungUpperLobe_L
LungUpperLobe_R	Lung_All	Lung_L
Lung_R	Musc_AnteriorScalene_L	Musc_AnteriorScalene_R
Nipple_L	Nipple_R	Process_Acromion
Process_Coracoid	RA	RCA
RV	Rib10_L	Rib10_R
Rib11_L	Rib11_R	Rib12_L
Rib12_R	Rib1_L	Rib1_R
Rib2_L	Rib2_R	Rib3_L
Rib3_R	Rib4_L	Rib4_R
Rib5_L	Rib5_R	Rib6_L
Rib6_R	Rib7_L	Rib7_R
Rib8_L	Rib8_R	Rib9_L
Rib9_R	Ribs	Scapula
Scapula_L	Scapula_R	SpinalCord

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Thorax — 97 (cont.)

SpinalCord_Ext	Sternum	SuperiorVenaCava
Trachea	V_Brachiocephalic_L	V_Brachiocephalic_R
V_Pulmonary	Vertebrae_T1	Vertebrae_T10
Vertebrae_T11	Vertebrae_T12	Vertebrae_T2
Vertebrae_T3	Vertebrae_T4	Vertebrae_T5
Vertebrae_T6	Vertebrae_T7	Vertebrae_T8
Vertebrae_T9		

Abdomen — 34

Bowel_Bag	Bowel_Bag_Small	Bowel_Small
Colon	Cyst_Kidney_L	Cyst_Kidney_R
Digestive	Duodenum	Fat_Visceral
GallBladder	Glnd_Adrenal_L	Glnd_Adrenal_R
Kidney_L	Kidney_R	Liver
LiverSegment_1	LiverSegment_2	LiverSegment_3
LiverSegment_4	LiverSegment_5	LiverSegment_6
LiverSegment_7	LiverSegment_8	Pancreas
Spine_Lumbar	Spleen	Stomach
V_Portal&Spleen	V_VenaCava_I	Vertebrae_L1
Vertebrae_L2	Vertebrae_L3	Vertebrae_L4
Vertebrae_L5		

Pelvis — 47

A_Iliac_L	A_Iliac_R	Bladder
Bone_Hip_L	Bone_Hip_R	Bone_Iliac_L
Bone_Iliac_R	Bone_Pelvic	Bone_Pubic
Bone_Sacrum	Canal_Anal	CaudaEquina
FemoralEpiphys	FemoralHead_E	FemurHead_withTro_L
FemurHead_withTro_R	Femur_Head_L	Femur_Head_R
Femur_L	Femur_R	GreaterTrochant
IliacCrest	IschialTuberosi	IschioPubicRam
LesserTrochante	Musc_GluteusMaximus_L	Musc_GluteusMaximus_R
Musc_GluteusMedius_L	Musc_GluteusMedius_R	Musc_GluteusMinimus_L
Musc_GluteusMinimus_R	Musc_Iliopsoas_L	Musc_Iliopsoas_R
PenileBulb	Prostate	PubisSymphysis
Rectum	Rectum_withAnal	SacroIliacCocc
SeminalVes	Sigmoid	Testis
TriradiateCarti	UteroCervix	V_Iliac_L
V_Iliac_R	Vertebrae_S1	

Nodal LN — 10

LN_Axillary_Level_III_L	LN_Axillary_Level_III_R	LN_Axillary_Level_II_L
LN_Axillary_Level_II_R	LN_Axillary_Level_I_L	LN_Axillary_Level_I_R
LN_IMN_L	LN_IMN_R	LN_Supraclavicular_L
LN_Supraclavicular_R		

Body — 9

Body	Fat_Subcutaneous	IntervertebralDiscs
Marrow	Muscles	SkeletalMuscle
Skeleton	Spine	VertebraeBody

Targets — 27

CTV	CTV-NL	CTV-NR
CTV1	CTV1_R	CTV1_With_Rib_R
CTV2	CTV2_L	CTV2_R

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Targets — 27 (cont.)

CTV_L	CTV_L_withRib	CTV_R
GTV	GTV-NL	GTV-NR
GTV-T	GTVnd(L)	GTVnd(R)
GTVnx	PCTV	PTV
PTV LN(L)	PTV LN(R)	PTV1
PTV2	PTVnx	pGTV

MRI — 91

AnalCanal	Aorta	Bladder
Body	BowelBag	BrachialPlexus_L
BrachialPlexus_R	Brain	BrainStem
Breast_L	Breast_R	BronchialTree
CTV	ChestWall_L	ChestWall_R
Cochlea_L	Cochlea_R	Colon
Digestive	Duodenum	Esophagus
Eye_L	Eye_R	FemurHead_withTro_L
FemurHead_withTro_R	GTV	GallBladder
Heart	InnerEar_L	InnerEar_R
Kidney_L	Kidney_R	Lacrimal_L
Lacrimal_R	Larynx	Lens_L
Lens_R	Lips	Liver
Lung_L	Lung_R	Mandible
Mandible_L	Mandible_R	Marrow
Mastoid_L	Mastoid_R	MiddleEar_L
MiddleEar_R	Muscles	Nipple_L
Nipple_R	OpticChiasm	OpticNerve_L
OpticNerve_R	OralCavity	OralCavity_withGum
PTV	Pancreas	Parotid_L
Parotid_R	PelvicBone	PenileBulb
PharyngealConstrictors_L	PharyngealConstrictors_M	PharyngealConstrictors_U
Pituitary	Prostate	Rectum
Rectum_withAnal	SeminalVesicle	Sigmoid
SkeletalMuscle	Skeleton	SmallBowel
SpinalCord	Spine	Spleen
Stomach	SubcutaneousFat	Submandibular_L
Submandibular_R	TMJ_L	TMJ_R
TemporalLobe_L	TemporalLobe_R	Testis
ThoracicFat	Thyroid	Trachea
VisceralFat		

Evidence & Limitations

Clinical Evidence: N/A

Limitations: Limited performance on contrast-enhanced CT scans; Reduced accuracy for post-surgical anatomy; Pediatric performance was not established in the reviewed direct AccuContour clinical validation studies; Requires manual verification and editing in complex anatomical regions; The structure library is the manufacturer-supplied list of 2026-09-23: 397 distinct CT ROI names (392 listed here, 5 unresolved vendor abbreviations omitted) and 91 MRI ROI names. A per-structure validation matrix is not published, so accuracy per structure cannot be verified.; MRI ROI models are listed by the manufacturer, while the cleared indication (FDA K251351) covers CT input; MRI use requires local validation.; Contouring guideline references are

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Limitations (cont.): supplied per structure by the manufacturer and are not independently verified; several structures are referenced to anatomical atlases rather than radiotherapy consensus guidelines.; The manufacturer-supplied Intertek MDR certificate lists AccuContour v3.1; it does not establish that FDA-cleared AccuContour 4.0 or MRI segmentation is within that EU version's scope.; K251351 compatible contouring input is non-contrast CT DICOM 3.0, including original CT and synthetic CT; contrast-enhanced or unusual imaging workflows require local validation.

Evidence by Source (9)

Establishing prospective performance monitoring for real-world implementation of deep learning-based auto-segmentation in prostate cancer radiotherapy

Zhu L et al. · Physics and Imaging in Radiation Oncology · 2025 · DOI 10.1016/j.phro.2025.100886

E1 L2 Sets product level

Single-centre prospective performance monitoring comparing two commercial systems in clinical use; geometric endpoints.

https://doi.org/10.1016/j.phro.2025.100886

Development and validation of a deep reinforcement learning algorithm for auto-delineation of organs at risk in cervical cancer radiotherapy

Yucheng L et al. · Scientific Reports · 2025 · DOI 10.1038/s41598-025-91362-9

E1 L2

Develops a reinforcement-learning refinement on top of AccuContour; geometric endpoints.

https://doi.org/10.1038/s41598-025-91362-9

Impact of Denoising on Deep-Learning-Based Automatic Segmentation Framework for Breast Cancer Radiotherapy Planning

Im JH et al. · Cancers (Basel) · 2022 · DOI 10.3390/cancers14153581

E1 L2

Single-centre study of denoising as input to AccuContour; geometric endpoints.

https://doi.org/10.3390/cancers14153581

Interobserver delineation variability of computed tomography-based radiomic features of the parotid gland

Buasawat K et al. · Radiat Oncol J · 2024 · DOI 10.3857/roj.2023.00605

E1 L3 Sets product level

Single-centre radiomic-feature variability study including AccuContour parotid contours.

https://doi.org/10.3857/roj.2023.00605

Online Adaptive Proton Therapy Facilitated by Artificial Intelligence-Based Autosegmentation in Pencil Beam Scanning Proton Therapy

Feng H et al. · Int J Radiat Oncol Biol Phys · 2025 · DOI 10.1016/j.ijrobp.2024.09.032

E1 L3

Single-centre online adaptive proton workflow facilitated by AccuContour; workflow endpoints.

https://doi.org/10.1016/j.ijrobp.2024.09.032

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AccuContour clinical evaluation

Wang J et al. · Front Oncol · 2022 · DOI 10.3389/fonc.2022.833816

E1 L2

Single-centre clinical evaluation with geometric endpoints.

https://doi.org/10.3389/fonc.2022.833816

Clinical validation of commercial deep-learning based auto-segmentation models for organs at risk in the head and neck region: a single institution study

Johnson CL et al. · Front. Oncol. · 2024 · DOI 10.3389/fonc.2024.1375096

E1 L2

Single-institution head-and-neck validation of two commercial systems for 22 OARs; AccuContour explicitly included.

https://doi.org/10.3389/fonc.2024.1375096

Consistency evaluation and performance optimization of deep learning-based auto-contouring for nasopharyngeal carcinoma

Yan L et al. · Sci Rep · 2025 · DOI 10.1038/s41598-025-33567-6

E1 L2

Single-centre consistency evaluation of four commercial models for nasopharyngeal structures.

https://doi.org/10.1038/s41598-025-33567-6

Patient-specific daily updated deep learning auto-segmentation for MRI-guided adaptive radiotherapy

Li Z, Zhang W, Li B, et al. · Radiother Oncol · 2022 · DOI 10.1016/j.radonc.2022.11.004

E1 L3

Six-patient longitudinal MRgART series with measured contouring and adaptive session times; Manteia co-authored.

https://doi.org/10.1016/j.radonc.2022.11.004

Regulatory Information

CE: cleared FDA: 510k_cleared

CE Certification:	cleared
CE Details:	cleared (Class Class IIa)
FDA Status:	510k_cleared
FDA Details:	510k_cleared (K251351)
Intended Use Statement:	AccuContour 4.0 is used by the radiation oncology department to segment CT images and generate information needed for treatment planning, treatment evaluation, and treatment adaption. (Source: FDA 510(k) K251351 Summary, accessed 2026-06-11.)

Market Information

Market Presence:	2020
On Market Since:	2020
Distribution Channels:	Direct sales; Partnerships

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Contact Information

Website:	https://www.manteiamedical.com/
Company URL:	https://www.manteiamedical.com/
Product URL:	https://www.manteiamedical.com/accucontour
Logo URL:	/logos/manteia.png
Logo Source:	Logo source: /logos/manteia.png
Support Email:	N/A

Quality Assurance

Last Revised:	9/29/2026
Company Revision Date:	N/A
Source:	FDA 510(k) database (K251351, K250780, K221706, K191928), Manteia official product page, peer-reviewed literature. 2026-09-23: manufacturer-supplied documentation added (vendor-provided, retrieved 2026-09-23) — 'AccuContour ROI full list' (411 model entries, 397 distinct TG-263 ROI names), 'MRI organs list' (88-ROI MRI model plus brain MRI target model), 'Technical Sheet: Contouring Guidelines in AccuContour' and a publication summary of 81 records. Intertek MDR Annex IX certificate 28620166008 and 2025-06-04 product list, manufacturer-supplied, received 2026-09-29.
GitHub URL:	https://github.com/DLinRT-eu/dlinrteu-website/tree/main/src/data/products/auto-contouring/manteia.ts

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